SPLIT PRIMES AND THE EXPONENT OF IMAGINARY QUADRATIC CLASS GROUPS

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Abstract. Assume that no Dirichlet L-function vanishes in the half-plane $\operatorname{Re} s > 11/12$, as a recent preprint asserts. Then every imaginary quadratic field $\mathbb{Q}(\sqrt{d})$ has a split prime below $(2\log|d|)^{12}$, its class group has exponent at least $\log(|d|/4)/(12\log(2\log|d|))$, and Euler's sixty-five idoneal numbers are all of them. With $\operatorname{Re} s > 7/8$ the exponent 12 becomes 8.

1. Introduction

Let $d < 0$ be a fundamental discriminant, $K = \mathbb{Q}(\sqrt{d})$, and let $E(d)$ be the exponent of the class group $\mathrm{Cl}(d)$ of K. Chowla [8] proved that only finitely many imaginary quadratic fields have a class group that is an elementary abelian 2-group, that is, $E(d) \le 2$; the proof is ineffective because of a possible Siegel zero. Under the generalised Riemann hypothesis, Boyd and Kisilevsky [6] and Weinberger [23] showed $E(d) \gg \log|d|/\log\log|d|$; without any hypothesis it is not known that $E(d) \to \infty$ [12]. Weinberger also showed that at most one field with $|d| > 5460$ has $E(d) \le 2$, and none under the generalised Riemann hypothesis. Heath-Brown [13] proved finiteness for the exponents 5, 2^r and $3 \cdot 2^r$ (see also Earnest and Körner [11]), and Elsenhans, Klüners and Nicolae [12] computed all fields with $E(d) \le 8$ and $|d| \le 3.1 \cdot 10^{20}$ and showed that their lists for $E(d) \in \{2, 4, 8\}$ are complete if no $L(s, \chi_d)$ has a real zero close to $s = 1$.[1]

All these results pass through one elementary observation. If a prime p splits in K as $p\mathcal{O}_K = \mathfrak{p}\bar{\mathfrak{p}}$ and $p^c < |d|/4$, then $\mathfrak{p}^c$ is not principal (Lemma 2.1). A split prime of size $(\log|d|)^{O(1)}$ therefore forces $E(d) \gg \log|d|/\log\log|d|$. Under the generalised Riemann hypothesis such a prime, of size $O(\log^2|d|)$, comes from the method of Ankeny [1], which Bach made explicit [2].

A zero-free half-plane is enough to replace the generalised Riemann hypothesis here. For $\frac{1}{2} \le B < 1$ write

$$\mathrm{H}(B): \qquad L(s,\chi) \neq 0 \quad \text{for every primitive Dirichlet character } \chi \text{ and every } s \text{ with } \operatorname{Re} s > B.$$

Two recent preprints of OpenAI assert H(7/8) [19] and H(11/12) [21]; a third excludes Landau–Siegel zeros uniformly [20]. These preprints have not been refereed. Every theorem below states the hypothesis it uses, and H(7/8) implies H(11/12).[2]

Theorem 1. *Assume* H(11/12). *Every imaginary quadratic field $\mathbb{Q}(\sqrt{d})$ has a prime $p \le (2\log|d|)^{12}$ that splits in it. Under* H(7/8) *the same holds with $p \le (2\log|d|)^8$.*

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[1] Their condition is that $L(s, \chi_d) \neq 0$ for $1 - 1/(4\log|d|) \le s < 1$ whenever $|d| > e^{11.2}$; it rests on Tatuzawa's explicit form of Siegel's theorem.

[2] The three preprints are dated 30 September, 5 October and 1 October 2026 and are kept in the repository https://github.com/openai/math.

Under H(B) the argument gives a split prime of size $(c\log|d|)^{1/(1-B)}$; the work lies in making c explicit and small and in covering all d at once, which Proposition 4.1 and Section 4 do with $c = 2$.[3]

Theorem 2. *Assume* H(7/8). *Every fundamental discriminant $d < -4$ satisfies*

$$E(d) \;\geq\; \frac{\log(|d|/4)}{8\log(2\log|d|)}.$$

Consequently every field with $E(d) \leq e$ has $|d| \leq N_e$, where $\log_{10} N_e$ is given in Table 1. Under H(11/12) *the same holds with* 12 *in place of* 8 *and the second row of the table.*

TABLE 1. Upper bounds $\log_{10} N_e$ for $|d|$ when $E(d) \leq e$.

e	2	3	4	5	6	7	8	9	10
H(7/8)	36.15	59.03	83.29	108.57	134.65	161.39	188.71	216.52	244.77
H(11/12)	59.03	95.82	134.65	174.99	216.52	259.05	302.42	346.53	391.30

The order $\log|d|/\log\log|d|$ is the one obtained from the generalised Riemann hypothesis; the half-plane only costs the factor $1 - B$ in the constant. For $e = 7$ even the finiteness of the set of fields with $E(d) = 7$ was not known without hypothesis [12]. Under H(11/12) the computation of Elsenhans, Klüners and Nicolae becomes unconditional for $E(d) \in \{2, 4, 8\}$ (Remark 4.2).

For exponent two we give a self-contained proof of the complete list. Let $D < 0$ be any discriminant and $C(D)$ the group of classes of primitive positive definite binary quadratic forms of discriminant D.

Theorem 3. *Assume* H(11/12). *Exactly* 101 *negative discriminants D have $C(D)$ of exponent at most* 2. *The largest is $|D| = 7392$; exactly* 65 *of them are fundamental, the largest being $|d| = 5460$. They are listed in Appendix C.*

A positive integer n is idoneal (Euler's *numerus idoneus*) if, roughly, every odd number properly represented by $x^2 + ny^2$ in only one way is prime; by Grube's theorem n is idoneal if and only if every genus of forms of discriminant $-4n$ consists of a single class [15], [9, Theorem 3.22 and Proposition 3.24]. Euler found 65 idoneal numbers, and without a hypothesis on zeros it is not known that there are no others [15].

Corollary 4. *Assume* H(11/12). *The idoneal numbers are exactly* 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 12, 13, 15, 16, 18, 21, 22, 24, 25, 28, 30, 33, 37, 40, 42, 45, 48, 57, 58, 60, 70, 72, 78, 85, 88, 93, 102, 105, 112, 120, 130, 133, 165, 168, 177, 190, 210, 232, 240, 253, 273, 280, 312, 330, 345, 357, 385, 408, 462, 520, 760, 840, 1320, 1365, 1848.

The completeness of Euler's list under H(11/12) was observed in [21], by combining the computation of [12] with Grube's reduction to fundamental discriminants. The proof given here is independent of both: it combines the lower bound $h(d) \geq 7612$ for $|d| > 2^{40}$ (Corollary 5.7), a sieve over $10^6 < |d| \leq 2^{40}$ by Lemma 2.1, a direct count of reduced forms for $|D| \leq 10^6$, and an arithmetic reduction of non-fundamental discriminants to fundamental ones (Section 6).

[3]The qualitative form, a prime with $\chi(p) = 1$ below a power of $\log q$ under a zero-free half-plane, is a classical consequence of the explicit formula. The preprint [21] records the analogous consequence for the least quadratic non-residue.

Method. Theorem 1 is proved with an explicit formula whose weight is a triangle in $\log n$ of half-width L centred at $\log y$ (Lemma 3.2). If no prime up to ye^L splits, the prime side is close to $-y$. Under $\mathrm{H}(B)$ each zero contributes at most y^B times a quantity controlled by the Hadamard product, and the sum over zeros is $O(y^B \log q)$. The choice $y \asymp (\log q)^{1/(1-B)}$ balances the two, and the inequalities of Rosser and Schoenfeld [22] make every constant explicit. Theorem 3 needs a lower bound for $L(1, \chi_d)$ instead, and Section 5 proves one by integrating a second explicit formula along the ray $s = 1 + u$, in the spirit of Littlewood [16]. All constants are certified in ball arithmetic [14].

Organisation. Section 2 collects the lemmas on split primes and forms. Section 3 proves the explicit formulas and bounds the zeros. Section 4 proves Theorems 1 and 2, Section 5 the lower bound for $L(1, \chi)$, and Section 6 Theorem 3. Appendix A contains the contour shifts, Appendix B the certified constants, Appendix C the list of Theorem 3.

2. Split primes, ideal classes and forms

Throughout, $d < 0$ is a fundamental discriminant, $K = \mathbb{Q}(\sqrt{d})$, $\chi_d = (\frac{d}{\cdot})$ is the Kronecker symbol, and $P(d)$ is the least prime that splits in K. A prime p splits in K if and only if $\chi_d(p) = 1$. The character χ_d is primitive of conductor $|d|$ and odd, $\chi_d(-1) = -1$ [10, Chapter 5].

Lemma 2.1 ([6, Lemma 2], [23, Lemma 5], [12, Lemma 1])**.** *Let p split in K, let $\mathfrak{p}$ be a prime of K above p, and let $c \geq 1$ be an integer with $p^c < |d|/4$. Then the order of the class of $\mathfrak{p}$ in $\mathrm{Cl}(d)$ exceeds c.*

Proof. Suppose $\mathfrak{p}^e = (\alpha)$ with $1 \leq e \leq c$. Write $\alpha = (x + y\sqrt{d})/2$ with $x, y \in \mathbb{Z}$. Then $p^e = N(\alpha) = (x^2 + |d|y^2)/4$. If $y \neq 0$ this is at least $|d|/4 > p^c \geq p^e$, which is impossible. Hence $y = 0$, $\alpha = x/2$ is a rational integer, and $x^2/4 = p^e$ forces e to be even and $(\alpha) = (p)^{e/2} = \mathfrak{p}^{e/2}\bar{\mathfrak{p}}^{e/2}$. Since p splits, $\mathfrak{p} \neq \bar{\mathfrak{p}}$, and unique factorisation of ideals contradicts $(\alpha) = \mathfrak{p}^e$. □

Corollary 2.2. *For $|d| > 4$, $E(d) \geq \log(|d|/4)/\log P(d)$.*

Proof. Let $p = P(d)$, $\mathfrak{p} \mid p$, and let m be the order of $\mathfrak{p}$. By Lemma 2.1 with $c = m$, $p^m \geq |d|/4$, so $E(d) \geq m \geq \log(|d|/4)/\log p$. □

The next bound is trivial but covers small discriminants.

Lemma 2.3. *$P(-3) = 7$, $P(-4) = 5$, and $P(d) \leq (|d| + 4)/4$ for every fundamental $d \leq -7$.*

Proof. The first two values are direct. If $d \equiv 1 \pmod 4$, put $q = |d|$ and $\alpha = (1 + \sqrt{d})/2$, of norm $(1+q)/4 > 1$. Let p be a prime factor of $(1+q)/4$. Then $p \nmid q$, since $p \mid 1+q$; and $p \nmid \alpha$ in $\mathcal{O}_K$, since $\alpha/p = (1/p + \sqrt{d}/p)/2 \notin \mathcal{O}_K$. A prime dividing the norm of an element that it does not divide, and not dividing d, splits. So $P(d) \leq (1+q)/4$. If $d = -4m$ with $m \equiv 1, 2 \pmod 4$ squarefree, the same argument applies to $\alpha = 1 + \sqrt{-m}$ of norm $1 + m$ and to an odd prime factor p of $1 + m$: such a prime exists, since $1 + m \equiv 3 \pmod 4$ when $m \equiv 2$, and $(1+m)/2$ is odd and exceeds 1 when $m \equiv 1 \pmod 4$, $m \geq 5$. It does not divide m, so it does not divide d. Hence $P(d) \leq 1 + m = (|d| + 4)/4$. □

For forms we follow Cox [9]. A form $ax^2 + bxy + cy^2$ of discriminant $D = b^2 - 4ac < 0$ with $a > 0$ is reduced if $|b| \leq a \leq c$, and $b \geq 0$ whenever $|b| = a$ or $a = c$. Every primitive positive definite form is properly equivalent to exactly one reduced form [9, Theorem 2.8], and a reduced form satisfies $a \leq \sqrt{|D|/3}$. The classes of primitive positive definite forms of discriminant D form a group $C(D)$ of order $h(D)$, isomorphic to the class group of the order of discriminant D in $\mathbb{Q}(\sqrt{D})$ [9, Theorem 7.7]; for fundamental $D = d$ it is $\mathrm{Cl}(d)$.

Lemma 2.4. *A reduced form (a,b,c) has order at most 2 in $C(D)$ if and only if $b=0$, $b=a$ or $a=c$.*

Proof. The inverse of the class of (a,b,c) is the class of $(a,-b,c)$. If $b=0$ the two forms coincide. If $b=a$ or $a=c$ (and then $b\geq 0$), the forms $(a,-b,c)$ and (a,b,c) are properly equivalent by $x\mapsto x+y$, $y\mapsto y$, respectively by $(x,y)\mapsto(-y,x)$. In all other cases $|b|<a<c$, so $(a,-b,c)$ is reduced and differs from (a,b,c), and by uniqueness of reduced forms the class differs from its inverse. □

Let $\mu(D)$ be defined as follows [9, Proposition 3.11]. Let r be the number of odd primes dividing D. If D is odd, $\mu(D)=r$. If $D=-4n$, then $\mu(D)=r$ when $n\equiv 3 \pmod 4$, $\mu(D)=r+1$ when $n\equiv 1,2 \pmod 4$ or $n\equiv 4 \pmod 8$, and $\mu(D)=r+2$ when $n\equiv 0 \pmod 8$. Genus theory gives $|C(D)/C(D)^2|=2^{\mu(D)-1}$ [9, Theorem 3.15], [7, Chapter 4].[4] Hence

$$C(D) \text{ has exponent at most } 2 \iff h(D)=2^{\mu(D)-1}. \tag{2.1}$$

For a fundamental discriminant d one checks from the definition that $\mu(d)=\omega(d)$, the number of distinct primes dividing d, so $E(d)\leq 2$ if and only if $h(d)=2^{\omega(d)-1}$.

3. Explicit formulas and the sum over zeros

Let χ be a primitive character modulo $q\geq 3$, $a\in\{0,1\}$ with $\chi(-1)=(-1)^a$, $\Lambda_\chi(n)=\Lambda(n)\chi(n)$, and let $\rho=\beta+i\gamma$ run over the nontrivial zeros of $L(s,\chi)$ with multiplicity. They satisfy $0<\beta<1$, the series $\sum_\rho|\rho|^{-2}$ converges, and the trivial zeros are $s=-2k-a$, $k\geq 0$ [10, Chapters 9 and 12]. Since ρ is a zero of $L(s,\chi)$ if and only if $1-\bar\rho$ is, H(B) implies $1-B\leq\beta\leq B$ for every ρ.

Lemma 3.1. *For $\sigma>1$,*

$$\sum_\rho \frac{\sigma-\beta}{(\sigma-\beta)^2+\gamma^2} \;\leq\; \frac12\log q+\frac12\psi\Big(\frac{\sigma+a}{2}\Big)-\frac12\log\pi-\frac{\zeta'}{\zeta}(\sigma),$$

where $\psi=\Gamma'/\Gamma$.

Proof. By the Hadamard product [10, Chapter 12],

$$\frac{L'}{L}(s,\chi)=-\frac12\log\frac{q}{\pi}-\frac12\psi\Big(\frac{s+a}{2}\Big)+b(\chi)+\sum_\rho\Big(\frac{1}{s-\rho}+\frac1\rho\Big),\qquad \operatorname{Re} b(\chi)=-\sum_\rho\operatorname{Re}\frac1\rho.$$

Taking real parts at $s=\sigma$ gives $\sum_\rho\operatorname{Re}\frac{1}{\sigma-\rho}=\frac12\log\frac{q}{\pi}+\frac12\psi(\frac{\sigma+a}{2})+\operatorname{Re}\frac{L'}{L}(\sigma,\chi)$, and $\operatorname{Re}\frac{1}{\sigma-\rho}=(\sigma-\beta)/|\sigma-\rho|^2$. Finally $|\frac{L'}{L}(\sigma,\chi)|=|\sum\Lambda_\chi(n)n^{-\sigma}|\leq-\frac{\zeta'}{\zeta}(\sigma)$. □

3.1. The triangle formula.

For $L>0$ and $y>1$ let

$$g(t)=\Big(1-\frac{|\log(t/y)|}{L}\Big)_+,\qquad F(s)=\int_{-L}^{L}\Big(1-\frac{|u|}{L}\Big)e^{su}\,du=\frac{4\sinh^2(sL/2)}{L\,s^2}.$$

The function g is continuous, supported on $[ye^{-L},ye^{L}]$, equal to 1 at $t=y$, and linear in $\log t$ on each side.[5]

[4] Cox states the count for $D=-4n$; for odd D the $\mu(D)$ assigned characters are the Legendre symbols of the primes dividing D, and the same proof applies [7].

[5] The Mellin transform of g is $\int_0^\infty g(t)t^{s-1}\,dt=y^sF(s)$, an entire function. The triangle is the convolution of two boxes of width L, which is why F is a square: $F(s)=L\big(\sinh(sL/2)/(sL/2)\big)^2$.

Lemma 3.2. *Let χ be odd, $L > 0$ and $y > e^L$. Then*

$$\sum_{n\geq 1} \Lambda_\chi(n)\, g(n) = -\sum_{\rho} y^{\rho} F(\rho) - \sum_{k\geq 0} y^{-1-2k} F(-1-2k),$$

where both sums converge absolutely.

Proof. By Mellin inversion, $g(t) = \frac{1}{2\pi i}\int_{(2)} y^s F(s) t^{-s}\, ds$, the integral converging absolutely because $|F(\sigma+it)| \leq 4\cosh^2(\sigma L/2)/(L|s|^2)$; here we used $|\sinh z|^2 = \sinh^2 \operatorname{Re} z + \sin^2 \operatorname{Im} z \leq \cosh^2 \operatorname{Re} z$. Since $-\frac{L'}{L}(s,\chi) = \sum \Lambda_\chi(n) n^{-s}$ converges absolutely on $\operatorname{Re} s = 2$,

$$\sum_{n} \Lambda_\chi(n) g(n) = \frac{1}{2\pi i}\int_{(2)} \Big(-\frac{L'}{L}(s,\chi)\Big) y^s F(s)\, ds.$$

The integrand is meromorphic with simple poles at the zeros of $L(s,\chi)$ only: $L(s,\chi)$ is entire, F is entire, and $L(0,\chi) \neq 0$ for odd χ. The residue at a zero ρ of multiplicity m is $-m\, y^\rho F(\rho)$, and at $s = -1-2k$ it is $-y^{-1-2k}F(-1-2k)$. Moving the line of integration to the left (Appendix A) proves the formula. For absolute convergence, $|y^\rho F(\rho)| \leq 4y\cosh^2(L/2)/(L|\rho|^2)$, and $|y^{-1-2k}F(-1-2k)| \leq (e^L/y)^{1+2k}/(L(1+2k)^2)$, which is summable because $y > e^L$. □

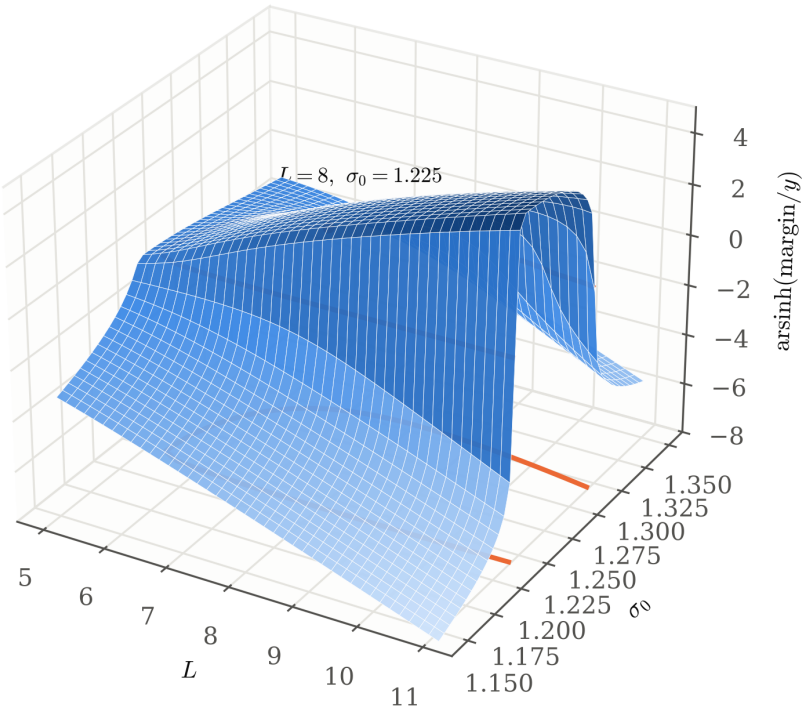


FIGURE 1. The certified quantity $(\Psi - R - 2E - T - Z)/y$ of Proposition 4.1 under H(7/8) at $\log q = 22$, with y chosen so that $ye^L = (2\log q)^8$, as a function of L and σ_0 (vertical scale arsinh). The orange curve on the floor is the boundary of the region where the quantity is positive; the point used in the proof is marked.

3.2. **The sum over zeros.**

Lemma 3.3. *Assume that every zero of $L(s,\chi)$ satisfies $1-B\le\beta\le B$, where $\frac12\le B<1$, let χ be odd, and let $\sigma_0>1$. Put*

$$h_0(\sigma_0)=\frac12\psi\Big(\frac{\sigma_0+1}{2}\Big)-\frac12\log\pi-\frac{\zeta'}{\zeta}(\sigma_0),\qquad M=\max\Big(4(\sigma_0-\tfrac12),\frac{1}{\sigma_0-B}\Big).$$

Then for $L>0$ and $y>1$,

$$\Big|\sum_\rho y^\rho F(\rho)\Big|\ \le\ Z:=\frac4L\Big(\frac12\log q+h_0(\sigma_0)\Big)\Big(y^B\cosh^2\frac{BL}{2}\cdot M+y^{1/2}\cosh^2\frac{L}{4}\cdot\frac{\sigma_0}{(1-B)^2}\Big).$$

Proof. Put $w(\rho)=(\sigma_0-\beta)/((\sigma_0-\beta)^2+\gamma^2)>0$. For every $R\ge\max((\sigma_0-\beta)/\beta^2,\,1/(\sigma_0-\beta))$ we have

$$(\sigma_0-\beta)^2+\gamma^2\le R(\sigma_0-\beta)\beta^2+R(\sigma_0-\beta)\gamma^2,\quad\text{that is,}\quad\frac{1}{|\rho|^2}\le R\,w(\rho).$$

If $\frac12\le\beta\le B$, then $(\sigma_0-\beta)/\beta^2\le4(\sigma_0-\frac12)$ and $1/(\sigma_0-\beta)\le1/(\sigma_0-B)$, so $R=M$ is allowed; also $y^\beta\cosh^2(\beta L/2)\le y^B\cosh^2(BL/2)$. If $1-B\le\beta<\frac12$, then $(\sigma_0-\beta)/\beta^2\le\sigma_0/(1-B)^2$ and $1/(\sigma_0-\beta)\le1/(\sigma_0-\frac12)<2\le\sigma_0/(1-B)^2$, because $(1-B)^2\le\frac14$ and $\sigma_0>1$; so $R=\sigma_0/(1-B)^2$ is allowed, and $y^\beta\cosh^2(\beta L/2)\le y^{1/2}\cosh^2(L/4)$. Since $|y^\rho F(\rho)|\le4y^\beta\cosh^2(\beta L/2)/(L|\rho|^2)$, each zero contributes at most $\frac4L\,w(\rho)$ times the bracket in Z. Summing over ρ and applying Lemma 3.1 with $a=1$ proves the lemma. □

4. The least split prime

Write $\theta(t)=\sum_{p\le t}\log p$ and $\psi(t)=\sum_{n\le t}\Lambda(n)$ (no confusion with the digamma function will arise). We use two inequalities of Rosser and Schoenfeld [22]:

$$\theta(t)>t\Big(1-\frac{1}{\log t}\Big)\quad(t\ge41),\qquad\psi(t)<1.03883\,t\quad(t>0),\tag{4.1}$$

which are [22, (3.16)] and [22, Theorem 12].

Proposition 4.1. *Let χ be a real, odd, primitive character modulo $q\ge3$ such that every zero of $L(s,\chi)$ satisfies $1-B\le\beta\le B$, where $\frac12\le B<1$. Let $L>0$, $y\ge41$ with $y>e^L$, put $Y=ye^L$, let $\sigma_0>1$, and let Z be as in Lemma 3.3. Put*

$$\Psi=\frac{y}{L}\Big[\Big(1-\frac{1}{\log y}\Big)(e^L-1)-1.03883\,(1-e^{-L})\Big],\qquad R=\log q\Big(1+\frac{2L}{\log2}\Big),$$

$$E=1.03883\Big(\sqrt{Y}+Y^{1/3}\frac{\log Y}{\log2}\Big),\qquad T=\frac{\pi^2e^L}{8Ly}.$$

If $\Psi-R-2E-T>Z$, then $\chi(p)=1$ for some prime $p\le Y$.

Proof. Suppose $\chi(p)\ne1$ for every prime $p\le Y$, and let $S=\sum_n\Lambda_\chi(n)g(n)$. Since χ is real, $\chi(p)\in\{-1,0\}$ for $p\le Y$, with $\chi(p)=0$ exactly when $p\mid q$.

Upper bound for S. The function g vanishes outside $[ye^{-L},Y]$ and $0\le g\le1$. For $p\nmid q$ we have $\chi(p)\Lambda(p)=-\Lambda(p)$ and $\chi(p^k)\Lambda(p^k)\le\Lambda(p^k)=-\Lambda(p^k)+2\Lambda(p^k)$ for $k\ge2$; for $p\mid q$ the term is $0=-\Lambda(p^k)+\Lambda(p^k)$. Hence

$$S\le-\sum_n\Lambda(n)g(n)+\sum_{p\mid q}\sum_k\Lambda(p^k)g(p^k)+2\sum_{\substack{p^k\le Y\\k\ge2}}\log p.$$

For $p \mid q$ at most $2L/\log p + 1$ powers p^k lie in $[ye^{-L}, Y]$, so the middle sum is at most $\sum_{p\mid q}(2L + \log p) \le 2L\omega(q) + \log q \le R$, because $\omega(q) \le \log q/\log 2$. The last sum is $\psi(Y) - \theta(Y) = \sum_{2\le k\le \log Y/\log 2} \theta(Y^{1/k})$, which by (4.1) is at most $1.03883\sqrt{Y} + (\log Y/\log 2)\, 1.03883\, Y^{1/3} = E$. For the first sum,

$$\sum_n \Lambda(n)g(n) \ge \sum_p \log p\, g(p) = -\int_{ye^{-L}}^{Y} \theta(t)g'(t)\,dt = \int_y^Y \frac{\theta(t)}{Lt}\,dt - \int_{ye^{-L}}^{y} \frac{\theta(t)}{Lt}\,dt,$$

by partial summation, since g is Lipschitz and vanishes at both ends, with $g'(t) = 1/(Lt)$ on (ye^{-L}, y) and $g'(t) = -1/(Lt)$ on (y, Y). By (4.1), the first integral is at least $(1 - 1/\log y)(Y - y)/L$ and the second is at most $1.03883\,(y - ye^{-L})/L$. Therefore $\sum_n \Lambda(n)g(n) \ge \Psi$ and $S \le -\Psi + R + 2E$.

Lower bound for S. By Lemma 3.2 and Lemma 3.3,

$$S \ge -\Big|\sum_\rho y^\rho F(\rho)\Big| - \sum_{k\ge 0} y^{-1-2k}F(-1-2k) \ge -Z - \frac{1}{L}\sum_{k\ge 0}\frac{(e^L/y)^{1+2k}}{(1+2k)^2} \ge -Z - T,$$

using $F(-1-2k) \le e^{(1+2k)L}/(L(1+2k)^2)$, $e^L/y < 1$ and $\sum_{k\ge 0}(2k+1)^{-2} = \pi^2/8$. Comparing the two bounds gives $\Psi - R - 2E - T \le Z$. □

Proof of Theorem 1. Let $B \in \{7/8, 11/12\}$, $k = 1/(1-B) \in \{8, 12\}$, $c = 2$, $L = 8$, and $\sigma_0 = 1.225$ for $B = 7/8$, $\sigma_0 = 1.27$ for $B = 11/12$. Let $q = |d|$ and $q_0 = e^{22}$.

Large discriminants. Let $q \ge q_0$ and choose $y = e^{-L}(c\log q)^k$, so that $Y = (c\log q)^k$. The character χ_d is real, odd and primitive modulo q, and $\mathrm{H}(B)$ gives $1 - B \le \beta \le B$. At $q = q_0$ the certified values (Appendix B) are $\log y = 22.2735\ldots$ and $(\Psi - R - 2E - T - Z)/y = 7.8330\ldots$ for $B = 7/8$, and $\log y = 37.4102\ldots$, $(\Psi - R - 2E - T - Z)/y = 12.0702\ldots$ for $B = 11/12$; in both cases $y \ge 41$, $y > e^L$ and $h_0(\sigma_0) > 0$. Figure 1 shows this quantity near the chosen parameters. We show that $(\Psi - R - 2E - T - Z)/y$ does not decrease as q grows; then Proposition 4.1 gives $P(d) \le Y = (2\log q)^k$ for all $q \ge q_0$.

As q grows, y grows. The quantity Ψ/y depends on q only through $1 - 1/\log y$, which increases. Next $R/y \propto \log q/(\log q)^k$ decreases, $T/y \propto y^{-2}$ decreases, $\sqrt{Y}/y = e^{L/2}y^{-1/2}$ decreases, and $Y^{1/3}\log Y/y = e^{L/3}y^{-2/3}\log(e^L y)$ decreases because $\log(e^L y) > 3/2$. In Z/y, the identity $(B-1)k = -1$ gives $y^{B-1} = e^{L/k}/(c\log q)$, so

$$\Big(\frac{1}{2}\log q + h_0\Big)y^{B-1} = \frac{e^{L/k}}{c}\Big(\frac{1}{2} + \frac{h_0}{\log q}\Big),$$

which decreases because $h_0 > 0$; and $(\frac{1}{2}\log q + h_0)y^{-1/2}$ is a constant times $(\frac{1}{2}t + h_0)t^{-k/2}$ with $t = \log q$, which decreases because $k/2 > 1$ and $h_0 > 0$. Hence every negative term of $(\Psi - R - 2E - T - Z)/y$ decreases in absolute value and the positive term increases.

Small discriminants. For $7 \le q \le q_0$ Lemma 2.3 gives $P(d) \le (q+4)/4$, and $(q+4)/4 < (2\log q)^8$ throughout this range. This is checked in interval arithmetic on the 412 intervals $[a, 1.05a]$ that cover $[7, q_0]$, comparing $(1.05a + 4)/4$ with $(2\log a)^8$; both sides increase, so this suffices. The same check holds with exponent 12. Finally $P(-3) = 7 < (2\log 3)^8$ and $P(-4) = 5 < (2\log 4)^8$. □

Proof of Theorem 2. By Corollary 2.2 and Theorem 1, $E(d) \ge \log(|d|/4)/\log P(d) \ge \log(|d|/4)/(k\log(2\log|d|))$. If $E(d) \le e$, then $\varphi(t) = t - \log 4 - ke\log(2t) \le 0$ at $t = \log|d|$. The function φ increases for $t > ke$ and tends to infinity, so $\log|d|$ is at most its largest zero t_e, and $N_e = e^{t_e}$. Table 1 lists $\log_{10} N_e$ rounded up, from a certified bisection. □

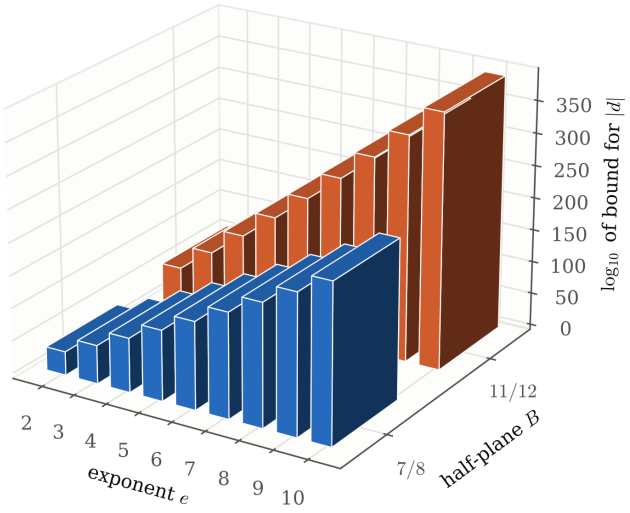


FIGURE 2. The bounds $\log_{10} N_e$ of Theorem 2: every imaginary quadratic field whose class group has exponent at most e has $|d| \le N_e$.

Remark 4.2. Elsenhans, Klüners and Nicolae show that if $L(s,\chi_d) \neq 0$ for $1-1/(4\log|d|) \le s < 1$ for every $|d| > e^{11.2}$, then $h(d) > 0.655\sqrt{|d|}/(\pi e \log|d|)$ for those d, and with this bound their computation finds every field with $E(d) \in \{2,4,8\}$ [12]. Since $1 - 1/(4\cdot 11.2) > 11/12$, the condition follows from H(11/12). Their lists of fields of exponent 4 and 8 are therefore complete under H(11/12), and for exponent 2 their list agrees with Theorem 3.

5. A lower bound for $L(1,\chi)$

The proof of Theorem 3 needs $h(d)$ to be large for $|d| > 2^{40}$. Theorem 2 alone is not enough: under H(7/8), $E(d) \le 2$ gives only $|d| \le 10^{36.15}$, far beyond the reach of a computation. We prove instead a lower bound for $L(1,\chi)$ of Littlewood's shape $1/\log\log q$. The notation is that of Section 3.

Lemma 5.1. *For $x > 1$ and $u \ge 0$,*

$$\sum_{n\le x} \frac{\Lambda_\chi(n)}{n^{1+u}} \log\frac{x}{n} = -\frac{L'}{L}(1+u,\chi)\log x - \Big(\frac{L'}{L}\Big)'(1+u,\chi) - \sum_{\rho} \frac{x^{\rho-1-u}}{(\rho-1-u)^2} - \sum_{k\ge 0} \frac{x^{-2k-a-1-u}}{(2k+a+1+u)^2},$$

and both sums converge absolutely.

Proof. Put $s = 1+u$. For $t > 0$ and $c > 0$, $\frac{1}{2\pi i}\int_{(c)} t^w w^{-2}\,dw = \max(\log t, 0)$ with absolute convergence, so termwise integration gives

$$\sum_{n\le x} \frac{\Lambda_\chi(n)}{n^s} \log\frac{x}{n} = \frac{1}{2\pi i}\int_{(1)} \Big(-\frac{L'}{L}(s+w,\chi)\Big)\frac{x^w}{w^2}\,dw.$$

The integrand has a double pole at $w=0$ with residue $-\frac{L'}{L}(s,\chi)\log x-(\frac{L'}{L})'(s,\chi)$, a pole at $w=\rho-s$ with residue $-x^{\rho-s}/(\rho-s)^2$ for each zero counted with multiplicity, and a pole at $w=-2k-a-s$ with residue $-x^{-2k-a-s}/(2k+a+s)^2$. Moving the line of integration to the left (Appendix A) proves the formula. Absolute convergence follows from $|\rho-s|\geq|\rho-1|\gg|\rho|$.[6] $\square$

Here $\log L(s,\chi)=\sum_{n\geq 2}\Lambda_\chi(n)n^{-s}/\log n$ for $\operatorname{Re}s>1$, continued along the real segment to $s=1$; this is possible because $L(1+u,\chi)\neq 0$ for $u\geq 0$.

Proposition 5.2. *For $x>1$ let $V(x)=\sum_{2\leq n\leq x}\frac{\Lambda_\chi(n)}{n\log n}\log\frac{x}{n}$. Then*

$$V(x)=\log L(1,\chi)\log x+\frac{L'}{L}(1,\chi)-\sum_{\rho}I_\rho(x)-T_1(x),$$

where

$$|I_\rho(x)|\leq\frac{x^{\beta-1}}{|\rho-1|^2\log x},\qquad |T_1(x)|\leq\frac{\pi^2}{8x\log x}.$$

Proof. Integrate Lemma 5.1 over $u\in[0,\infty)$. On the left, $\int_0^\infty n^{-1-u}\,du=1/(n\log n)$ gives $V(x)$. On the right, $\int_0^\infty-\frac{L'}{L}(1+u,\chi)\,du=\log L(1,\chi)$ and $\int_0^\infty-(\frac{L'}{L})'(1+u,\chi)\,du=\frac{L'}{L}(1,\chi)$, since $\log L(1+u,\chi)$ and $\frac{L'}{L}(1+u,\chi)$ tend to 0. The zero term is $I_\rho(x)=\int_0^\infty x^{\rho-1-u}(\rho-1-u)^{-2}\,du$; since $|\rho-1-u|^2=(1-\beta+u)^2+\gamma^2\geq|\rho-1|^2$,

$$|I_\rho(x)|\leq\frac{x^{\beta-1}}{|\rho-1|^2}\int_0^\infty x^{-u}\,du=\frac{x^{\beta-1}}{|\rho-1|^2\log x}.$$

In the same way the k-th trivial term is at most $x^{-2k-a-1}/((2k+a+1)^2\log x)$, and $\sum_{k\geq 0}(2k+a+1)^{-2}\leq\pi^2/8$. The interchanges of sum and integral are justified by these bounds and $\sum_\rho|\rho-1|^{-2}<\infty$. $\square$

Corollary 5.3. *Let $1<y<x$, $\ell=\log(x/y)$, and let $w(n)=1$ for $n\leq y$, $w(n)=\log(x/n)/\ell$ for $y<n\leq x$, $w(n)=0$ for $n>x$. Then*

$$\log L(1,\chi)=\sum_{n\geq 2}\frac{\Lambda_\chi(n)w(n)}{n\log n}+\vartheta\,\mathcal{E},\qquad|\vartheta|\leq 1,$$

where, with $\beta^*=\sup_\rho\beta$,

$$\mathcal{E}=\frac{1}{\ell}\Big[\Big(\frac{x^{\beta^*-1}}{\log x}+\frac{y^{\beta^*-1}}{\log y}\Big)\sum_\rho\frac{1}{|\rho-1|^2}+\frac{\pi^2}{8}\Big(\frac{1}{x\log x}+\frac{1}{y\log y}\Big)\Big].$$

Proof. For $n\leq y$, $\log(x/n)-\log(y/n)=\ell$, and for $y<n\leq x$ only $\log(x/n)$ occurs, so $V(x)-V(y)=\ell\sum\Lambda_\chi(n)w(n)/(n\log n)$. Subtracting the formulas of Proposition 5.2 at x and y removes $\frac{L'}{L}(1,\chi)$, and the bounds there give the claim. $\square$

Lemma 5.4. *If every zero of $L(s,\chi)$ has $\beta\leq B$, where $\frac12\leq B<1$, then for every $u>0$*

$$\sum_\rho\frac{1}{|\rho-1|^2}\leq S(q,u,B):=\frac{1+u-B}{(1-B)^2}\Big(\frac12\log\frac{q}{\pi}+\frac12\psi\Big(1+\frac{u}{2}\Big)-\frac{\zeta'}{\zeta}(1+u)\Big).$$

[6] The square in the kernel x^w/w^2 is what makes the sum over zeros converge absolutely without truncation in height.

Proof. Let $\sigma = 1+u$. Since $\sigma-\beta \ge 1-\beta > 0$, we have $|\sigma-\rho| \le \frac{\sigma-\beta}{1-\beta}|1-\rho|$, hence

$$\frac{1}{|1-\rho|^2} \le \frac{\sigma-\beta}{(1-\beta)^2}\cdot\frac{\sigma-\beta}{|\sigma-\rho|^2}.$$

The function $\beta \mapsto (\sigma-\beta)/(1-\beta)^2$ has derivative $(2\sigma-1-\beta)/(1-\beta)^3 > 0$ on $(-\infty,1)$, so it is at most $(\sigma-B)/(1-B)^2$. Sum over ρ and apply Lemma 3.1, using $\psi((\sigma+a)/2) \le \psi(1+u/2)$. □

Let $B_1 = 0.2614972128\ldots$ be Mertens' constant and γ Euler's constant, and put

$$G(y) = \sum_{p\le y}\Big(\frac{1}{p} - \log\Big(1+\frac{1}{p}\Big)\Big), \qquad W = \frac{1}{\ell}\int_{\log y}^{\log x}\Big(\log v + B_1 + \frac{1}{2v^2}\Big)dv.$$

The terms of G are positive, so G increases.

Lemma 5.5. *Let $286 \le y < x$, $\ell = \log(x/y)$, and let $\mathcal{M}$ be the sum in Corollary 5.3. Then*

$$-W + G(y) - \frac{3}{\sqrt{y}} \le \operatorname{Re}\mathcal{M} \le W + \gamma - B_1 + \frac{3}{\sqrt{y}}.$$

Proof. Since $\Lambda(p^k)/(p^k\log p^k) = 1/(kp^k)$ and $w(p^k) = 1$ for $p^k \le y$,

$$\mathcal{M} = \sum_{p\le y} -\log\Big(1-\frac{\chi(p)}{p}\Big) + \sum_{y<p\le x}\frac{\chi(p)w(p)}{p} + \mathcal{R},$$

where

$$\mathcal{R} = \sum_{\substack{p\le y,\ k\ge 2\\ p^k>y}}\frac{\chi(p)^k(w(p^k)-1)}{kp^k} + \sum_{\substack{p>y\\ k\ge 2}}\frac{\chi(p)^k w(p^k)}{kp^k}.$$

As $0\le w\le 1$, $|\mathcal{R}| \le \frac12\sum_p\sum_{k\ge2,\,p^k>y}p^{-k}$. For fixed p let k_0 be the least $k\ge2$ with $p^k>y$; the inner sum is at most $p^{-k_0}p/(p-1) \le 2p^{-k_0}$. If $p\le\sqrt{y}$ then $p^{-k_0} < 1/y$, and there are at most $\sqrt{y}$ such primes; if $p>\sqrt{y}$ then $k_0=2$. Hence $|\mathcal{R}| \le 1/\sqrt{y} + \sum_{m>\sqrt{y}}m^{-2} \le 1/\sqrt{y} + 1/(\sqrt{y}-1) \le 3/\sqrt{y}$.

For the lower bound, $-\log|1-\chi(p)/p| \ge -\log(1+1/p)$ and $\operatorname{Re}\chi(p)w(p)/p \ge -w(p)/p$. With $P(t) = \sum_{p\le t}1/p$, partial summation gives the identity

$$\sum_{p\le y}\frac{1}{p} + \sum_{y<p\le x}\frac{\log(x/p)}{\ell\,p} = \frac{1}{\ell}\int_y^x P(t)\,\frac{dt}{t},$$

and Rosser and Schoenfeld [22, (3.18)] give $P(t) < \log\log t + B_1 + 1/(2\log^2 t)$ for $t\ge 286$. With $v=\log t$ the right-hand side is at most W, which proves the lower bound. For the upper bound we use $-\log|1-\chi(p)/p| \le -\log(1-1/p)$ and Mertens' identity

$$\sum_p\Big(-\log\Big(1-\frac{1}{p}\Big) - \frac{1}{p}\Big) = \gamma - B_1,$$

whose terms are positive, so that $\sum_{p\le y} -\log(1-1/p) \le P(y) + \gamma - B_1$.[7] The rest is as before. □

[7]The identity follows from Mertens' theorems $\prod_{p\le t}(1-1/p)^{-1} \sim e^\gamma\log t$ and $\sum_{p\le t}1/p = \log\log t + B_1 + o(1)$ by taking logarithms [18, Theorem 2.7].

Theorem 5.6. *Assume* $\mathrm{H}(B)$ *with* $\frac12 \le B < 1$ *and put* $k = 1/(1-B)$. *Let* $q_0 \ge 3$, $y_0 \ge 10^6$, $\ell > 0$, $u > 0$, *and suppose that* $c_0 = \frac12\psi(1+\frac{u}{2}) - \frac12\log\pi - \frac{\zeta'}{\zeta}(1+u) > 0$. *Let*

$$E_0 = \frac{1}{\ell}\Big[\Big(\frac{x_0^{B-1}}{\log x_0} + \frac{y_0^{B-1}}{\log y_0}\Big)S(q_0,u,B) + \frac{\pi^2}{8}\Big(\frac{1}{x_0\log x_0} + \frac{1}{y_0\log y_0}\Big)\Big], \qquad x_0 = e^{\ell}y_0,$$

$\kappa_0 = \log y_0 + \ell/2 - k\log\log q_0$, *and*

$$C_- = \exp\Big(G(10^6) - B_1 - \frac{1}{2\log^2 y_0} - \frac{3}{\sqrt{y_0}} - E_0\Big).$$

Then every primitive character χ *of conductor* $q \ge q_0$ *satisfies* $|L(1,\chi)| \ge C_-/(k\log\log q + \kappa_0)$.

Proof. Take $y = y_0(\log q/\log q_0)^k$ and $x = e^{\ell}y$ in Corollary 5.3, so that $y \ge y_0$ and $\log y + \ell/2 = k\log\log q + \kappa_0$. Under $\mathrm{H}(B)$, $\beta^* \le B$, and Lemma 5.4 bounds the zero sum by $S(q,u,B) = A(\frac12\log q + c_0)$ with $A = (1+u-B)/(1-B)^2$. Since $(B-1)k = -1$,

$$y^{B-1}S(q,u,B) = y_0^{B-1}\log q_0 \cdot A\Big(\frac12 + \frac{c_0}{\log q}\Big),$$

which decreases in q because $c_0 > 0$; the same holds with x in place of y, and the remaining factors of $\mathcal{E}$ decrease too. Hence $\mathcal{E} \le E_0$. For the prime sum, concavity of log gives $\ell^{-1}\int_{\log y}^{\log x}\log v\,dv \le \log(\log y + \ell/2)$, and $\ell^{-1}\int_{\log y}^{\log x} dv/(2v^2) \le 1/(2\log^2 y_0)$, so $W \le \log(k\log\log q + \kappa_0) + B_1 + 1/(2\log^2 y_0)$. Also $G(y) \ge G(10^6)$ and $3/\sqrt{y} \le 3/\sqrt{y_0}$. Taking real parts in Corollary 5.3 and inserting Lemma 5.5 gives $\log|L(1,\chi)| \ge -W + G(y) - 3/\sqrt{y} - E_0$, and exponentiating proves the theorem. □

For a fundamental $d < -4$ the class number formula reads $h(d) = \sqrt{|d|}\,L(1,\chi_d)/\pi$ [10, Chapter 6].

Corollary 5.7. *Let* $d < -2^{40}$ *be fundamental. Under* $\mathrm{H}(7/8)$,

$$h(d) \;\ge\; \frac{0.72554\sqrt{|d|}}{\pi\,(8\log\log|d| - 4.15398)} \;\ge\; 10\,798.96,$$

and under $\mathrm{H}(11/12)$, $h(d) \ge 0.71408\sqrt{|d|}/(\pi(12\log\log|d| - 8.55734)) \ge 7\,612.06$.

Proof. Apply Theorem 5.6 with $q_0 = 2^{40}$ and, for $B = 7/8$, $y_0 = 5\cdot10^6$, $\ell = 14$, $u = 1/10$; for $B = 11/12$, $y_0 = 1.8\cdot10^9$, $\ell = 20$, $u = 0.08$. The certified constants are in Appendix B. The function $f(q) = C_-\sqrt{q}/(\pi(k\log\log q + \kappa_0))$ increases for $q \ge 2^{40}$, since $\frac{d}{dq}\log f(q) = \frac{1}{2q} - \frac{k}{q\log q\,(k\log\log q + \kappa_0)} > 0$ there, and $f(2^{40})$ is the stated number. □

6. Class groups of exponent two

Proof of Theorem 3. We first determine the fundamental discriminants d with $E(d) \le 2$, in three ranges, and then pass to all discriminants.

$|d| \le 10^6$. For every discriminant D with $|D| \le 10^6$ we listed the primitive reduced forms (a,b,c), using $a \le \sqrt{|D|/3}$, and checked by Lemma 2.4 whether all of them have order at most 2. Exactly 101 discriminants pass, 65 of them fundamental, the largest $|D|$ being 7392 and the largest fundamental $|d|$ being 5460.[8]

$10^6 < |d| \le 2^{40}$. If $E(d) \le 2$, Lemma 2.1 with $c = 2$ shows that no prime p with $4p^2 < |d|$ splits in K. In particular 2 does not split, so $d \not\equiv 1 \pmod 8$: either $d = -q$ with $q \equiv 3$

[8]For $|D| \le 10^6$ the enumeration covers non-fundamental discriminants as well, which gives an independent check of the reduction carried out below.

(mod 8), or $d = -4m$ with $m \equiv 1, 2 \pmod 4$. For every such q, respectively m, in the range we tested the condition $\chi_d(p) \neq 1$ for the odd primes $p < 1000$ with $4p^2 < |d|$. A wheel modulo $8 \cdot 3 \cdot 5 \cdots 23$, respectively $4 \cdot 3 \cdot 5 \cdots 23$, discards the residue classes that fail at some $p \leq 23$; this leaves 1 088 640 classes in each case and about $1.34 \cdot 10^9$ values of q, respectively m, to test. None passes. Squarefreeness was not used, so the conclusion covers every fundamental d in the range.[9]

$|d| > 2^{40}$. Suppose $E(d) \leq 2$, so that $h(d) = 2^{\omega(d)-1}$. Let $t = 15$ under H(7/8) and $t = 14$ under H(11/12). If $\omega(d) < t$, then $h(d) \leq 2^{t-2}$, which is $8192 < 10\,798.96$, respectively $4096 < 7\,612.06$, contradicting Corollary 5.7. If $\omega(d) = s \geq t$, then $|d| \geq p_s\#$, the product of the first s primes, and $p_s\# \geq p_t\# > 2^{40}$. Let f be the increasing function of the proof of Corollary 5.7. We claim $f(p_s\#) > 2^{s-1}$ for all $s \geq t$, which contradicts $h(d) \geq f(|d|) \geq f(p_s\#)$. For $s = t$ the certified values are $f(p_{15}\#) = 7.088\ldots \cdot 10^6 > 2^{14}$ and $f(p_{14}\#) = 7.468\ldots \cdot 10^5 > 2^{13}$ in the two cases. For the step from s to $s+1$, write $q = p_s\#$ and $p = p_{s+1} \leq q$. Then $\log\log(qp) \leq \log(2\log q) = \log\log q + \log 2$, so with $A_q = k\log\log q + \kappa_0$,

$$\frac{f(qp)}{f(q)} \;\geq\; \sqrt{p}\,\frac{A_q}{A_q + k\log 2} \;\geq\; \sqrt{p_{t+1}}\,\frac{A_{2^{40}}}{A_{2^{40}} + k\log 2} \;\geq\; 2,$$

because $A_q \geq A_{2^{40}} > 0$ for $q > 2^{40}$, the factor $A/(A + k\log 2)$ increases with A, and $p \geq p_{t+1}$, which is $p_{16} = 53$, respectively $p_{15} = 47$; the certified values $A_{2^{40}} = 22.42\ldots$, respectively $31.30\ldots$, give $\sqrt{p_{t+1}}\,A_{2^{40}}/(A_{2^{40}} + k\log 2) \geq 2$ in both cases. Hence $f(p_{s+1}\#) \geq 2f(p_s\#) > 2^s$.

So the fundamental discriminants with $E(d) \leq 2$ are the 65 found in the first range.

Non-fundamental discriminants. Let $D = f^2 d$ with $f > 1$ and d fundamental, and suppose $C(D)$ has exponent at most 2. The natural map $C(D) \to \mathrm{Cl}(d)$ is surjective [9, §7], so $E(d) \leq 2$ and d is one of the 65. By the class number formula for orders [9, Theorem 7.24],

$$h(D) = \frac{h(d)\,f}{u}\prod_{p\mid f}\Big(1 - \frac{\chi_d(p)}{p}\Big), \qquad u = [\mathcal{O}_K^\times : \mathcal{O}^\times] \in \{1, 2, 3\},$$

with $u = 3$ only for $d = -3$ and $u = 2$ only for $d = -4$. By (2.1), $h(D) = 2^{\mu(D)-1}$ and $h(d) = 2^{\omega(d)-1}$. Every odd prime dividing D divides d or f, and comparing the definition of μ in the cases $2 \nmid D$, $2 \mid f$, $2 \mid d$ shows $\mu(D) \leq \omega(d) + \omega(f) + 1$. Since $\prod_{p\mid f}(1 - \chi_d(p)/p) \geq \prod_{p\mid f}(1 - 1/p)$, we get

$$\frac{\varphi(f)}{3} \leq \frac{h(D)}{h(d)} \leq 2^{\omega(f)+1}, \qquad \text{so} \qquad \prod_{p^a \parallel f}\frac{\varphi(p^a)}{2} \leq 6.$$

Each factor $\varphi(p^a)/2 = p^{a-1}(p-1)/2$ is at least 1, except $\varphi(2)/2 = \frac{1}{2}$, so each factor is at most 12. This forces $p \leq 23$ (for $p \geq 29$ the factor is at least 14), and $2^a \mid f$ with $a \leq 5$, 3^a with $a \leq 3$, 5^a with $a \leq 2$, and $a \leq 1$ for $7 \leq p \leq 23$. Hence f divides $F = 2^5 \cdot 3^3 \cdot 5^2 \cdot 7 \cdot 11 \cdot 13 \cdot 17 \cdot 19 \cdot 23$, which has 4608 divisors. For each of the 65 fundamental d and each divisor f of F we evaluated $h(D)$ by the class number formula and compared it with $2^{\mu(D)-1}$. Equality holds for exactly 36 pairs with $f > 1$; together with the 65 fundamental discriminants they are the 101 discriminants found in the first range, all with $|D| \leq 7392$. □

Proof of Corollary 4. By Grube's theorem, in the form of [9, Theorem 3.22 and Proposition 3.24], n is idoneal if and only if $h(-4n) = 2^{\mu(-4n)-1}$, that is, by (2.1), if and only if $C(-4n)$ has exponent at most 2. The discriminants $D \equiv 0 \pmod 4$ among the 101 of Theorem 3 are $-4n$ for exactly the 65 numbers listed. □

[9] This range is also covered by the unconditional tabulation of the class groups of imaginary quadratic fields with $|d| < 2^{40}$ by Mosunov and Jacobson [17]; the sieve above does not use it.

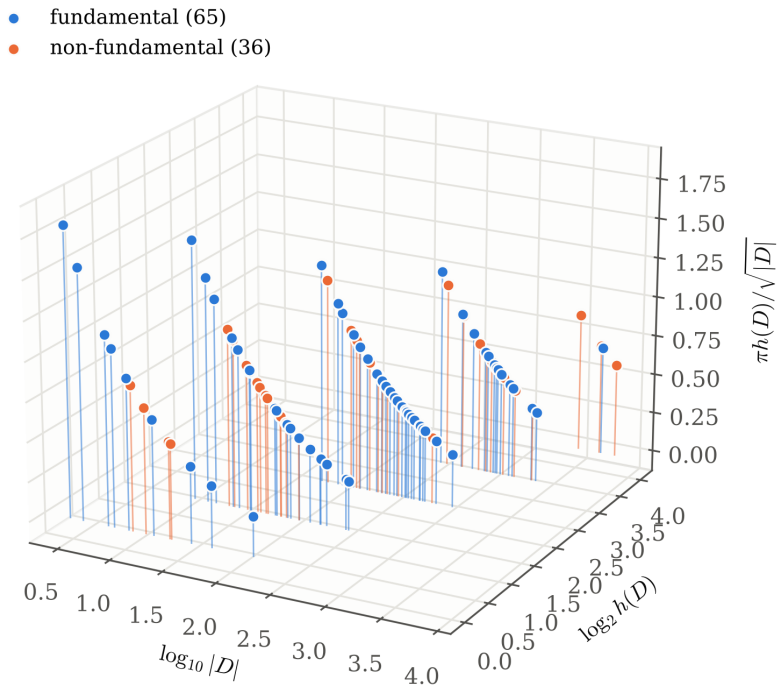


FIGURE 3. The 101 discriminants of Theorem 3: $\log_{10}|D|$, $\log_2 h(D)$ and $\pi h(D)/\sqrt{|D|}$, which is $L(1,\chi_d)$ for fundamental $d<-4$. The fields of exponent two have small $L(1,\chi_d)$, because every small prime is inert or ramified; this is the tension that Corollary 5.7 resolves.

7. Concluding remarks

The exponent $k=1/(1-B)$ in Theorems 1 and 2 is the only place where the width of the zero-free region enters. Under the generalised Riemann hypothesis the same argument has $k=2$. Any improvement of the half-plane, or a zero-density estimate near $\operatorname{Re} s=B$, improves the constants at once. The method uses only the real character χ_d; the same proof gives, for every real primitive character of conductor q, a prime $p\le(2\log q)^k$ with $\chi(p)=1$ when χ is odd. The half-plane $\operatorname{Re} s>7/8$ also yields the Pólya–Vinogradov inequality with $\log\log q$ [5], and explicit prime computations of the kind used here underlie the author's bound for Lehmer's totient problem [4]. Questions about the exponent of a finitely generated group arise in a different setting in the author's work on hyperbolic groups [3].

Without a hypothesis on zeros, at most one fundamental discriminant is missing from Theorem 3 [23], and if it exists it has $|d|>2^{40}$ by the sieve of Section 6.[10] A refereed proof of $\mathrm{H}(B)$ for any $B<1$ would remove the exception and make Theorems 1, 2 and 3 unconditional, with $k=1/(1-B)$ in place of 8 or 12 after the certification of Appendix B is repeated.

[10]The sieve uses only Lemma 2.1, so this part is unconditional.

Appendix A. The contour shifts

We justify the shifts of contour in Lemmas 3.2 and 5.1. In both, the integrand is $-\frac{L'}{L}(s',\chi)\Phi(s')$ with s' the argument of L: $\Phi(s') = y^{s'}F(s')$ in Lemma 3.2, and $\Phi(s') = x^{s'-s}/(s'-s)^2$ with $s = 1+u$ in Lemma 5.1. In both cases there are $\theta > 1$ and C such that

$$|\Phi(\sigma+it)| \le C\,\theta^{\sigma}\,(1+|t|)^{-2} \qquad (\sigma \le 2,\ |\sigma+it-s_*| \ge 1), \tag{A.1}$$

where $s_* = 0$, respectively $s_* = s$: take $\theta = ye^{-L}$ for $\sigma \le 0$ and $\theta = y$ for $0 \le \sigma \le 2$ in the first case, since $|F(\sigma+it)| \le 4\cosh^2(\sigma L/2)/(L|\sigma+it|^2) \le 4e^{|\sigma|L}/(L|\sigma+it|^2)$; and $\theta = x$ in the second.[11] We use three facts [10, Chapters 12, 16, 17 and 19], together with $|\frac{L'}{L}(\sigma+it,\chi)| \le -\frac{\zeta'}{\zeta}(2)$ for $\sigma \ge 2$.

(1) For each $j \ge 2$ there is $T_j \in [j, j+1]$ such that $|\frac{L'}{L}(\sigma \pm iT_j,\chi)| \ll \log^2(qT_j)$ uniformly for $-1 \le \sigma \le 2$.
(2) For $\sigma \le -1$ and s' at distance at least $\frac12$ from every trivial zero, $|\frac{L'}{L}(s',\chi)| \ll \log(q|s'|)$, by the functional equation and Stirling's formula.
(3) The number of zeros with $|\gamma - t| \le 1$ is $O(\log(q(|t|+2)))$.

Let $m \ge 1$ and let $\mathcal{R}$ be the rectangle with vertices $2 \pm iT_j$ and $-U_m \pm iT_j$, where $U_m = 2m+a+1$, so that the left side lies midway between two trivial zeros. By the residue theorem the integral over the right side equals the sum of the residues inside $\mathcal{R}$ plus the integrals over the other three sides. On the horizontal sides, (A.1) and (1), (2) bound the integrand by $\ll \theta^{\sigma}\log^2(q(T_j+U_m+2))/T_j^2$, so their contribution is $\ll \theta^2\log^2(q(T_j+U_m+2))/(T_j^2\log\theta) \to 0$ as $j \to \infty$ with m fixed. On the left side, (2) gives the bound $\ll \theta^{-U_m}\int_{-\infty}^{\infty}\log(q(U_m+|t|+2))(1+|t|)^{-2}\,dt \ll \theta^{-U_m}\log(qU_m) \to 0$ as $m \to \infty$. Letting first $j \to \infty$ and then $m \to \infty$, and using (3) to see that the residues at the nontrivial zeros form an absolutely convergent series, we obtain the two formulas. In Lemma 5.1 the right side of $\mathcal{R}$ is the line $\operatorname{Re} s' = 1+s$ instead of $\operatorname{Re} s' = 2$, which changes nothing.

Appendix B. Certified constants

All constants were computed with the ball arithmetic of Arb [14] at 256 bits, through python-flint; every digit shown is guaranteed. The values $\zeta'/\zeta(\sigma)$ come from the Arb power series of ζ, the values of ψ from its digamma function, and $G(10^6) = 0.181981816529021\ldots$ is a sum of 78 498 terms in balls. Mertens' constant was taken to fifty digits.

Theorem 1	$B = 7/8$	$B = 11/12$
$k,\ L,\ \sigma_0,\ c,\ \log q_0$	8, 8, 1.225, 2, 22	12, 8, 1.27, 2, 22
$\log y$ at q_0	22.273517…	37.410276…
$h_0(\sigma_0)$	3.1316109…	2.4139115…
M	2.9	3.08
Ψ/y	355.64128…	362.40792…
Z/y	347.80610…	350.33768…
$(R+2E+T)/y$	0.0021171…	0.0000009…
$(\Psi-R-2E-T-Z)/y$	7.8330627…	12.070238…

[11] In the first case two values of θ are used on the two sides of $\sigma = 0$; on the left of the contour only $\theta = ye^{-L} > 1$ matters.

Corollary 5.7	$B = 7/8$	$B = 11/12$
$q_0,\ y_0,\ \ell,\ u$	$2^{40},\ 5 \cdot 10^6,\ 14,\ 0.1$	$2^{40},\ 1.8 \cdot 10^9,\ 20,\ 0.08$
$S(q_0, u, B)$	$323.75077\ldots$	$587.32747\ldots$
c_0	$8.6197489\ldots$	$11.108462\ldots$
E_0	$0.2378758\ldots$	$0.2560657\ldots$
κ_0	$-4.1539838\ldots$	$-8.5573459\ldots$
C_-	$0.7255434\ldots$	$0.7140853\ldots$
$f(2^{40})$	$10\,798.962\ldots$	$7\,612.065\ldots$
$f(p_t\#)$	$7.0888 \cdot 10^6$ $(t = 15)$	$7.4688 \cdot 10^5$ $(t = 14)$

Appendix C. The discriminants of exponent at most two

The 101 discriminants of Theorem 3, given as $|D|$ and grouped by $h(D)$; the fundamental ones are in bold.

$h = 1$**: 3**, **4**, **7**, **8**, **11**, 12, 16, **19**, 27, 28, **43**, **67**, **163**.
$h = 2$**: 15**, **20**, **24**, 32, **35**, 36, **40**, 48, **51**, **52**, 60, 64, 72, 75, **88**, **91**, 99, 100, 112, **115**, **123**, 147, **148**, **187**, **232**, **235**, **267**, **403**, **427**.
$h = 4$**: 84**, 96, **120**, **132**, 160, **168**, 180, 192, **195**, **228**, 240, **280**, 288, **312**, 315, **340**, 352, **372**, **408**, **435**, 448, **483**, **520**, **532**, **555**, **595**, **627**, **708**, **715**, **760**, **795**, **928**, **1012**, **1435**.
$h = 8$**: 420**, 480, **660**, 672, **840**, 960, **1092**, 1120, **1155**, 1248, **1320**, **1380**, **1428**, **1540**, **1632**, **1848**, **1995**, **2080**, **3003**, **3040**, **3315**.
$h = 16$**: 3360**, **5280**, **5460**, 7392.

Declarations

Data availability. The certification script, the sieve, the enumeration of reduced forms, the reduction of non-fundamental discriminants and their outputs are archived in [24]. The sieve and the enumeration are C programs; the other computations use Python with Arb through python-flint and SymPy.

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Ethics statement. This study involves no human participants, animals or personal data; no ethical approval was required.

Use of AI tools. Claude Code (Opus 5.5 and Fable 5.1, Anthropic), GPT-6.1 Sol and GPT-6 Astra (OpenAI) were used to write and run the programs behind the computations and to prepare the manuscript. The hypotheses H(7/8) and H(11/12) are results claimed in preprints of OpenAI [19, 21], cited as such. The author checked every statement and is responsible for the content.

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